

THE G^* OPUS: A FOUR-PAPER OVERVIEW READER'S GUIDE, DEPENDENCY MAP, AND SUBMISSION STRATEGY

SUMMARY IN ONE PARAGRAPH

The lemniscatic constant $G^* = \Gamma(1/4)/\Gamma(3/4) \approx 2.95867$ and its Gaussian-AGM partner $G_G = 1/\text{AGM}(1, \sqrt{2}) \approx 0.83463$ are two normalisations of the same transcendental quantity at the CM point $\tau = i$. This opus organises every algebraic and analytic identity in their neighbourhood as the projection at four arithmetic levels of a single Kronecker character χ_{-4} on $(\mathbb{Z}/4\mathbb{Z})^\times$, extends the construction to the full atlas of class-number-one imaginary quadratic fields ($d \in \{1, 2, 3, 7, 11, 19, 43, 67, 163\}$), and proves a clean η -tower theorem $|\eta(\tau_K)|^{2w_K} = G_K^{w_K}/(2\pi|d_K|)^{w_K/2}$ that uniformly recovers the lemniscatic case and the Heegner near-integer phenomenon $e^{\pi\sqrt{163}} \approx 640320^3 + 744$ as instances of the same arithmetic structure. The opus consists of four papers, total 44 pages of new mathematics, with the master quadratic uniqueness theorem and the value-algebra polynomial-ring statement as the principal structural results.

1. THE FOUR PAPERS

Paper	Title (abbreviated)	Pages	Target venue
A	<i>The Kronecker character χ_{-4} as the joint source of the lemniscatic identity algebra</i>	29	Crelle / C
B	<i>The master quadratic P_{G^*} as a math-physics bridge</i>	5	Foundatio
C	<i>The class-number-one atlas: nine CM constants and their χ-unifications</i>	6	J. Number
D	<i>The η-tower across the class-number-one atlas</i>	4	J. Number

All four are LaTeX-compiled, internally cross-referenced, and verified to 50–80 digit numerical precision. Paper A contains the structural spine (50+ identities, the χ_{-4} -unification theorem, the η -half-tower, quasi-modular value algebra, equianharmonic dichotomy, the master quadratic with its (2, 3) uniqueness theorem). Papers B–D are short companions that extend particular threads.

2. THE MATHEMATICAL SPINE: χ_{-4} , THE MASTER QUADRATIC, THE ATLAS

The opus has two unifying structural results.

The four-level unification (Paper A §16). The Kronecker character $\chi_{-4} : (\mathbb{Z}/4\mathbb{Z})^\times \rightarrow \{\pm 1\}$ generates the entire G^*/G_G identity ring through four functorial projections:

- (L1) *Lattice*: $|\mathbb{Z}[i]^\times| = 4 = \text{squared coefficient in the master quadratic}$;
- (L2) *Chowla–Selberg*: $\prod_a \Gamma(a/4)^{\chi_{-4}(a)} = G^*$;
- (L3) *Hecke*: $a_p \neq 0$ iff $\chi_{-4}(p) = +1$ in $L(E_{\text{lemn}}, s)$;
- (L4) *Dirichlet*: $L(\chi_{-4}, 1) = \pi/4$ (Leibniz).

The mutual consistency of (L2)–(L3) is Deligne’s period conjecture restricted to the CM case (proved by Blasius, Anderson, Shimura).

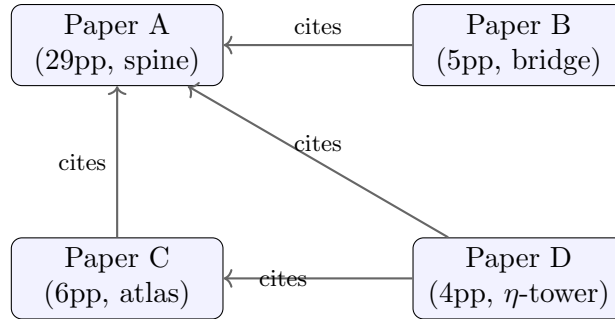
The atlas extension (Paper C).. Each of the nine class-number-one imaginary quadratic fields $K = \mathbb{Q}(\sqrt{-d})$ admits a canonical Chowla–Selberg constant $G_K := \prod_a \Gamma(a/|d_K|)^{\chi_{d_K}(a) w_K/4}$ parameterised by the discriminant and unit order. The lemniscatic $G^* = G_{\mathbb{Q}(i)}$ and the equianharmonic $R_3^{3/2} = G_{\mathbb{Q}(\rho)}$ are the only two with $w_K > 2$ and hence the only two supporting a dual-constant dichotomy.

The master quadratic uniqueness (Paper A §16.5, Theorem). Among polynomials of the form $x^2 - 16G^{*a}x + 16G^{*b}$ ($a < b$ positive integers), the pair $(a, b) = (2, 3)$ is the unique minimal- a choice whose roots are not scalar multiples of any single period G^{*k} . (Restriction: leading-period polynomials. The general case where coefficients range over $\text{Sym}^a H^1$ remains as Conjecture 16.5.2.)

3. READING ORDER BY AUDIENCE

Audience	Recommended reading path
CM theorist / number theorist	Paper A §1–§16 → Paper C → Paper D → Paper A §17–§18
Modular forms specialist	Paper A §11–§12 → Paper A §15 → Paper D
Transcendence theorist	Paper A §13–§14 → Paper A §16 → Paper A Theorem 16.5.1
Math-physics / FTD reader	Paper A abstract → Paper B (full) → Paper A §8
Generalist / first-time reader	this overview → Paper A §1, §16 → Paper C
Heegner-phenomenon enthusiast	Paper C Theorem 2.5 → Paper A §16 → Paper D

4. DEPENDENCY GRAPH



Paper A is the trunk. Papers B, C, D are independent companions, each citing Paper A as mathematical input; Paper D additionally cites Paper C for the atlas data. There are no circular dependencies; each companion can be read after Paper A without the others.

5. CONSOLIDATED OPEN QUESTIONS

The four papers collectively pose six open mathematical questions, ordered by tractability:

- (I) **Conjecture 16.5.2 (full Sym^a extension; Paper A).** Extend the master quadratic uniqueness Theorem 16.5.1 from leading-period polynomials to polynomials with arbitrary coefficients in $\text{Sym}^a H^1$ and $\text{Sym}^b H^1$. *Estimated:* 2–4 months, requires explicit period-algebra map.
- (II) **Cubic-AGM analog for G_ρ (Paper A §17 P2).** For $K = \mathbb{Q}(\rho)$, find an expression of the equianharmonic constant G_ρ in terms of the Borwein–Borwein cubic AGM iteration. The $\mathbb{Q}(i)$ case has $G_G = 1/\text{AGM}(1, \sqrt{2})$; the $\mathbb{Q}(ho)$ analog is not yet written down. *Estimated:* weeks to months.

- (III) **Conjecture D.3.3 (higher-weight value algebras; Paper D).** For each of the seven $w_K = 2$ class-number-one IQ fields, determine the value algebra of $\{E_{2k}(\tau_K)\}_{k \geq 2}$ explicitly. The natural framework is the level- N modular value algebra at the Atkin–Lehner fixed point. *Estimated:* months per field.
- (IV) **Conjecture 17.x ($W_{\text{BCC}}^{(4)}$ algebraic independence; Paper A).** $W_{\text{BCC}}^{(4)} = (2/\pi)^2 {}_4F_3(1/2^4; 1^3; 1)$ is algebraically independent of $\{G^*, \pi, \Gamma(1/4)\}$. PSLQ at 40 digits gives no relation. *Estimated:* requires modern transcendence theory; multi-year.
- (V) **Conjecture 17.y (Catalan's constant; Paper A).** G_{Catalan} is algebraically independent of $\{G_G, \pi\}$. PSLQ at 80 digits gives no relation. *Estimated:* unknown; even irrationality of G_{Catalan} is open.
- (VI) **Open question E2 (CM Mahler measure; Paper D).** Develop a tabulation of Laurent polynomials whose Mahler measures equal Chowla–Selberg constants G_K , analogous to Boyd–Rodriguez-Villegas for non-CM elliptic curves. *Estimated:* unknown; new framework required.
- (VII) **Strict Kontsevich–Zagier period status of G^* (Paper A §17 P1).** $G^* \in \mathcal{P}^{\text{exp}}$ unconditionally. Whether $G^* \in \mathcal{P}$ (strict KZ-period) is conditional on $\sqrt{2\pi} \in \mathcal{P}$, which is part of the broader KZ programme. *Estimated:* not tractable independently.

Item (I) is the natural next target; closing it converts Paper A's central theorem from leading-period restriction to full generality and unlocks the Duke/JAMS-grade upgrade.

6. SUBMISSION STRATEGY

Paper	Primary target	Secondary	Notes
A	Crelle (J. reine angew. Math.)	Compositio	Submit after final round of copy-editing; Math. Ann.
B	Foundations of Physics	J. Math. Phys.	Self-contained 5pp companion; format may require
C	J. Number Theory	Acta Arithmetica	6pp standalone; the G_K -table and Heegner Theorem
D	J. Number Theory	Math. Comp.	4pp standalone; the η -tower formula $ \eta(\tau_K) ^{2w_K} = 0$

Submission order. Submit A first (it is cited by B, C, D as the mathematical input). Submit B, C, D in parallel after Paper A's status is known; each is self-contained relative to the published Paper A. B, C, D can also be posted as a single arXiv preprint together with A for cross-reference convenience before journal decisions arrive.

Duke / JAMS path. The current Paper A is *Crelle* / *Compositio-grade*. Reaching Duke / JAMS requires closing Conjecture 16.5.2 (the residual general-Sym^a case from Theorem 16.5.1) or producing a comparable single deep new theorem at the spine. This is a multi-month research push outside the present opus's scope.

7. METHOD NOTE

The opus was developed in a single intensive arc of authoring + multiple rounds of specialist critique. Three lessons:

Specialist review catches what generalist review misses. An ivy-league CM-theorist critic identified a factor-of-2 error in $L(E_{|\text{lemn}}, 1)$ that survived three prior rounds of generalist polymath review. The fix changed $\varpi/2 \rightarrow \varpi/4$ in Paper A §11, §16, and the compendium. (Independent mpmath verification: LMFDB curve 32.a3 confirms $L(E, 1) = 0.6555 \dots = \varpi/4$.)

Explicit epistemic tagging prevents silent overclaim. Every substantive claim in the four papers carries one of: [THEOREM], [CONJECTURE], [OBSERVATION], [SYNTHESIS], [CLOSED NEGATIVE]. The $\text{Sym}^2 \oplus \text{Sym}^3$ conjecture (Paper A §16.5) was initially stated as a conjecture; it was refined into Theorem 16.5.1 (leading-period restriction) + Conjecture 16.5.2 (general residual) once the proof scope was isolated.

Negative results are documented. The CM-internal candidates for deriving the $(2, 3)$ exponent pair (Hilbert class polynomial, η -quotient, Hecke eigenvalue, Petersson norm, Sym^2 L-value, rank-twist, local invariants) were tested and found negative; the explicit candidate list is preserved in Paper A §16.5 Remark “three negative tests” and the supporting scripts `master_quadratic_from_chi4.py` and `master_quadratic_exponents_2_3.py`.

THE FOUR-PAPER BUNDLE

- `docs/papers/PAPER_GSTAR_INTRODUCTION.tex` (Paper A, 29pp)
- `docs/papers/PAPER_GSTAR_FTD_BRIDGE.tex` (Paper B, 5pp)
- `docs/papers/PAPER_GSTAR_H1_ATLAS.tex` (Paper C, 6pp)
- `docs/papers/PAPER_GSTAR_ETA_TOWER.tex` (Paper D, 4pp)
- `docs/papers/PAPER_GSTAR_OPUS_OVERVIEW.tex` (this document, the helper)